

QuickSort in Python: Ekdum Simple Guide!



Hello Soubhik! Python aur DSA seekhna sach mein bahut majedar ho sakta hai. Yeh concept itna aasan hai ki ise bhaiya ke ladke ko bhi khel-khel mein sikhaya jaa sakta hai. Aaj hum QuickSort ko step-by-step tod kar samjhenge.

1. QuickSort Kya Hota Hai? 🤔

QuickSort ek **Divide and Conquer** (Baanto aur Raj karo) algorithm hai. Iska basic matlab hai ki ek badi problem ko chhote-chhote hisso mein tod do aur fir solve karo.

Real-Life Example: Socho gym mein workout ke baad bahut saari alag-alag weight ki plates bikhri padi hain. Aap ek 10kg ki plate ko uthate ho aur usko standard (Pivot) maan lete ho. Ab saari halki plates ek taraf (left), aur bhari wali plates dusri taraf (right) karte ho. Bas, isi tarah QuickSort data ko sort karta hai!

2. 'Pivot' Ka Magic ✨

QuickSort mein hum ek **Pivot** (Leader) chunte hain. Aam taur par array ka pehla element hamara Pivot ban jata hai.

Pivot ka ek hi maqsad hota hai: **Apni sahi jagah par jaana**. Sahi jagah matlab, ek aisi position jiske left wale saare elements usse chhote (ya barabar) hon, aur right wale saare elements usse bade hon.

Maan lo ek nayi shipment aayi hai brass lights ki. Agar unhe size ke hisaab se sort karna hai, toh sabse pehli light ko apna 'Pivot' maan kar baaki saari lights ko easily do hisso mein baanta jaa sakta hai.

Diagram: Pivot in Action

Hamara array hai: [22, 35, 20, 15, 40]. Chalo 22 ko Pivot banate hain.

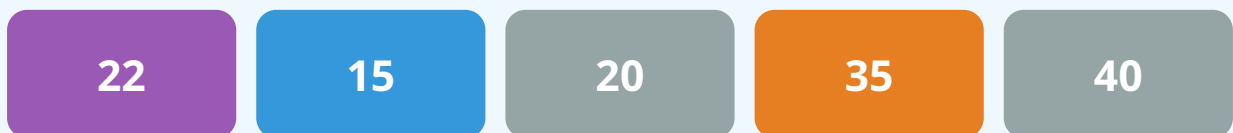


Humein 22 ko array ke beech mein wahan rakhna hai, jahaan chote elements left mein aur bade right mein hon.

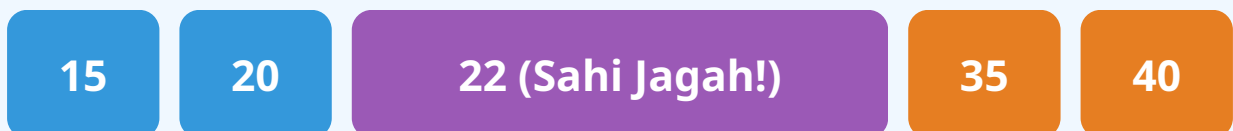
3. Step-by-Step Example

Hum do point set karenge: **i** (Left se right jayega) aur **j** (Right se left aayega).

- **Step 1:** Pivot = 22. 'i' dhoondega Pivot se **BADA** number. 'j' dhoondega Pivot se **CHHOTA** number.
- **Step 2:** 'i' aage badha toh usko mila **35**. 'j' pichhe aaya toh usko mila **15**.
- **Step 3:** Dono (35 aur 15) apni galat jagah par hain! Toh hum inko **SWAP (badal)** kar denge.



- **Step 4:** Ab 'i' aur aage badhega aur 'j' pichhe aayega. Jab 'i' aur 'j' ek dusre ko cross kar jayenge, toh process ruk jayega.
- **Step 5:** Aakhir mein hum **Pivot (22) ko 'j' wale element se swap** kar denge.



Hooray! 22 apni sahi jagah par aa gaya. Ab left wale [15, 20] aur right wale [35, 40] ko alag-alag same tarike se (Recursion) sort karenge. Yeh ekdum backend (Django) mein bade datasets handle karne jaisa logic hai.

4. Python Code (Ekdum Halwa! 🍲)

UrbanPro par agar kisi student ko Python sikhana ho, toh yeh simple code snippet best rahega:

```
def partition(arr, l, r):
    pivot = arr[l] # Pehla element leader (pivot) hai
    i = l + 1
    j = r

    while True:
        # 'i' ko aage badhao jab tak chhota ya barabar element mile
        while i <= j and arr[i] <= pivot:
            i += 1
        # 'j' ko pichhe lao jab tak bada ya barabar element mile
        while i <= j and arr[j] >= pivot:
            j -= 1

        if i <= j:
            # Galat jagah wale elements ko swap karo
            arr[i], arr[j] = arr[j], arr[i]
        else:
            break # i aur j cross ho gaye!

    # Pivot ko uski aakhiri sahi jagah (j) par daal do
    arr[l], arr[j] = arr[j], arr[l]
    return j # Sahi index wapas bhejo

def quick_sort(arr, l, r):
    if l < r:
        p = partition(arr, l, r) # Pivot ko set karo
        quick_sort(arr, l, p - 1) # Left hisse ko sort karo
        quick_sort(arr, p + 1, r) # Right hisse ko sort karo

# Chalo Test Karte Hain!
my_array = [22, 35, 20, 15, 40]
quick_sort(my_array, 0, len(my_array) - 1)
print(my_array) # Output: [15, 20, 22, 35, 40]
```

5. Important Points Yaad Rakhne Ke Liye 💡

- **Time Complexity:**
 - **Average Case:** $O(N \log N)$ - Yeh normal halat mein kafi fast kaam karta hai, ekdum Rapido ride ki tarah!
 - **Worst Case:** $O(N^2)$ - Agar data pehle se hi sorted ho ya reverse sorted ho, tab yeh algorithm thoda phas jata hai aur extra time leta hai.
- **Merge Sort vs Quick Sort:** Merge Sort hamesha $O(N \log N)$ time leta hai, lekin usme memory (space) zyada lagti hai. QuickSort in-place hota hai isliye memory bachata hai.

Keep Coding! Data Analyst aur GenAI safar ke liye All the Best!